



**King Abdul Aziz University**  
**Faculty of Computing & Information Technology**  
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## **CPCS 212 - APPLIED MATH FOR COMPUTING**

# **Nonlinear Equation**

# **Newton-Raphson Method**

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# Newton-Raphson Method

(Also known as Newton's Method)

Given an initial guess of the root  $\mathbf{x}_0$ , Newton-Raphson method uses information about the function and its derivative at that point to find a better guess of the root.

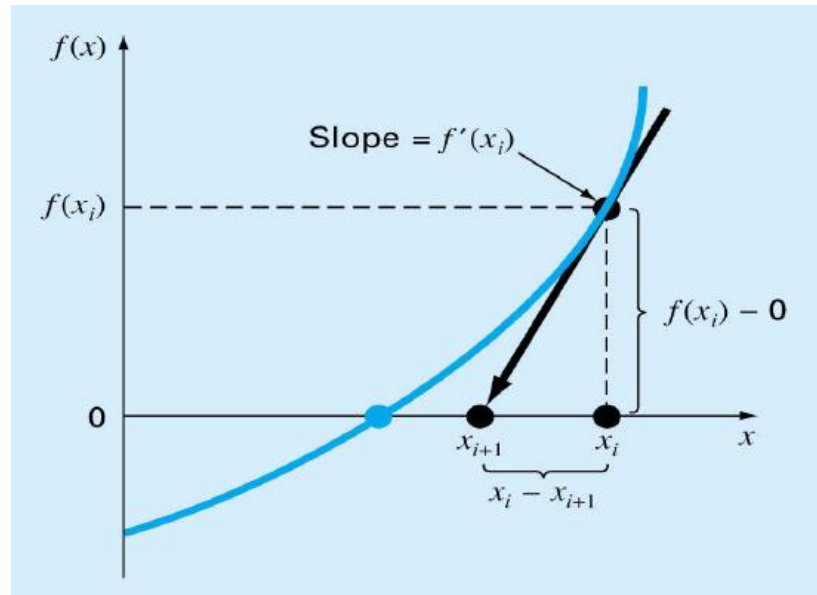
## Assumptions:

- $f(\mathbf{x})$  is continuous and the first derivative is known
- An initial guess  $\mathbf{x}_0$  such that  $f'(\mathbf{x}_0) \neq 0$  is given

# Newton Raphson Method

## - Graphical Depiction -

- If the initial guess at the root is  $x_i$ , then a tangent to the function of  $x_i$  that is  $f'(x_i)$  is extrapolated down to the  $x$ -axis to provide an estimate of the root at  $x_{i+1}$ .



# Derivation of Newton's Method

*Given:*  $x_i$  an initial guess of the root of  $f(x) = 0$

*Question:* How do we obtain a better estimate  $x_{i+1}$ ?

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Taylor Theorem:  $f(x+h) \approx f(x) + f'(x)h$

Find  $h$  such that  $f(x+h) = 0$ .

$$\Rightarrow h \approx - \frac{f(x)}{f'(x)}$$

Newton – Raphson Formula

A new guess of the root:  $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$

# Newton's Method

*Given*  $f(x)$ ,  $f'(x)$ ,  $x_0$

*Assumption*  $f'(x_0) \neq 0$

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*for*  $i = 0:n$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

*end*

*C    FORTRAN PROGRAM*

$F(X) = X^{**3} - 3 * X^{**2} + 1$

$FP(X) = 3 * X^{**2} - 6 * X$

$X = 4$

*DO* 10  $I = 1, 5$

$X = X - F(X) / FP(X)$

*PRINT* \*,  $X$

10    *CONTINUE*

*STOP*

*END*

# Newton's Method

Given  $f(x)$ ,  $f'(x)$ ,  $x_0$

Assumption  $f'(x_0) \neq 0$

---

for  $i = 0:n$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

end

```
function [F]=F(X)
```

$$F = X^3 - 3 * X^2 + 1$$

```
function [FP]=FP(X)
```

$$FP = 3 * X^2 - 6 * X$$

```
% MATLAB PROGRAM
```

```
X = 4
```

```
for i = 1:5
```

$$X = X - F(X) / FP(X)$$

```
end
```



# Example

Find a zero of the function  $f(x) = x^3 - 2x^2 + x - 3$  ,  $x_0 = 4$

$$f'(x) = 3x^2 - 4x + 1$$

$$\text{Iteration 1:} \quad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 4 - \frac{33}{33} = 3$$

$$\text{Iteration 2:} \quad x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 3 - \frac{9}{16} = 2.4375$$

$$\text{Iteration 3:} \quad x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.4375 - \frac{2.0369}{9.0742} = 2.2130$$



# Example

| <b>k<br/>(Iteration)</b> | <b><math>x_k</math></b> | <b><math>f(x_k)</math></b> | <b><math>f'(x_k)</math></b> | <b><math>x_{k+1}</math></b> | <b><math> x_{k+1} - x_k </math></b> |
|--------------------------|-------------------------|----------------------------|-----------------------------|-----------------------------|-------------------------------------|
| 0                        | 4                       | 33                         | 33                          | 3                           | 1                                   |
| 1                        | 3                       | 9                          | 16                          | 2.4375                      | 0.5625                              |
| 2                        | 2.4375                  | 2.0369                     | 9.0742                      | 2.2130                      | 0.2245                              |
| 3                        | 2.2130                  | 0.2564                     | 6.8404                      | 2.1756                      | 0.0384                              |
| 4                        | 2.1756                  | 0.0065                     | 6.4969                      | 2.1746                      | 0.0010                              |

# Example

For  $f(x) = x^3 - 4$ , perform 3 iterations of Newton's method with starting point  $x_0 = 2$ .

# Convergence Analysis

Theorem :

Let  $f(x)$ ,  $f'(x)$  and  $f''(x)$  be continuous at  $x \approx r$  where  $f(r) = 0$ . If  $f'(r) \neq 0$  then there exists  $\delta > 0$

such that  $|x_0 - r| \leq \delta \Rightarrow \frac{|x_{k+1} - r|}{|x_k - r|^2} \leq C$

$$C = \frac{1}{2} \frac{\max_{|x_0 - r| \leq \delta} |f''(x)|}{\min_{|x_0 - r| \leq \delta} |f'(x)|}$$

# Convergence Analysis

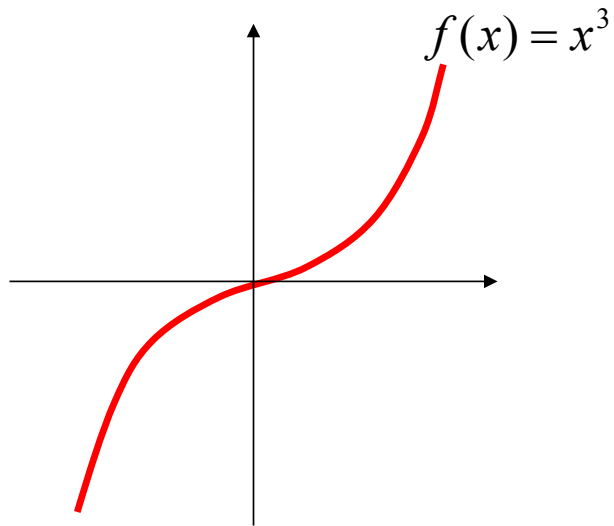
## Remarks

- When the guess is close enough to a **simple** root of the function then Newton's method is guaranteed to converge quadratically.
- Quadratic convergence means that the number of correct digits is nearly doubled at each iteration.

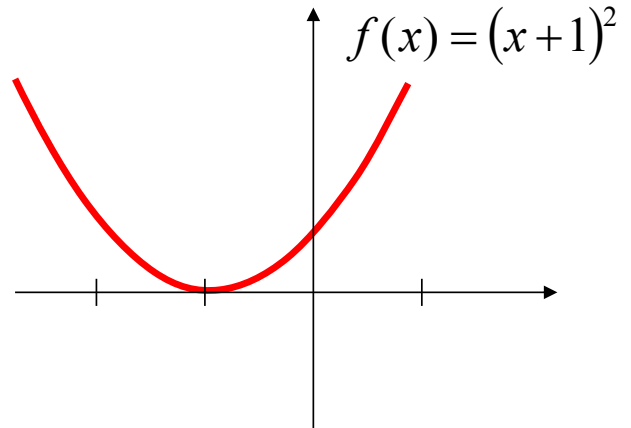
# Problems with Newton's Method

- If the initial guess of the root is far from the root the method may not converge.
- Newton's method converges linearly near multiple zeros  $\{ f(r) = f'(r) = 0 \}$ . In such a case, modified algorithms can be used to regain the quadratic convergence.

# Multiple Roots



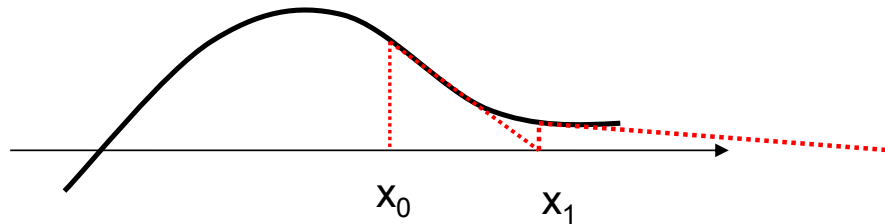
$f(x)$  has three  
zeros at  $x = 0$



$f(x)$  has two  
zeros at  $x = -1$

# Problems with Newton's Method

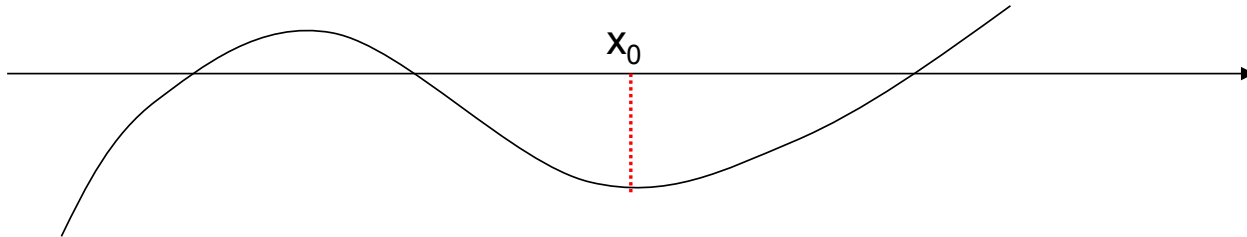
## - Runaway -



The estimates of the root is going away from the root.

# Problems with Newton's Method

## - Flat Spot -



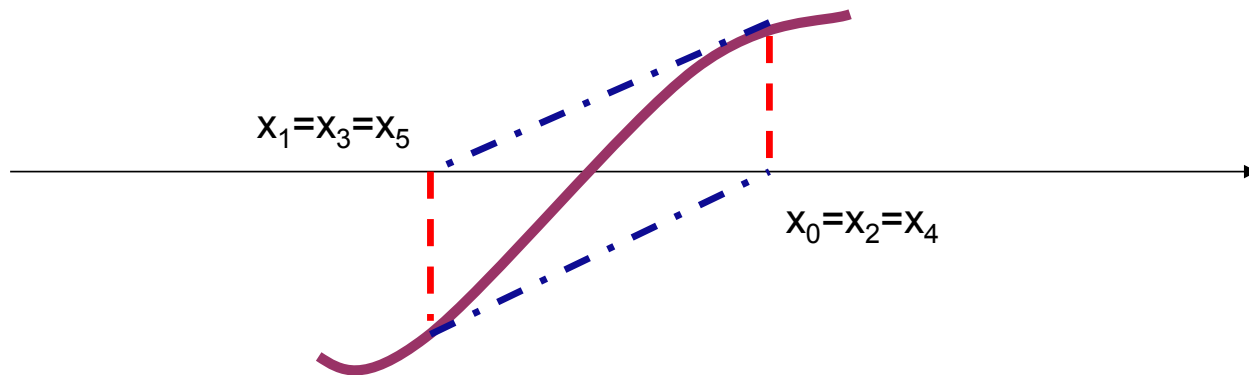
The value of  $f'(x)$  is zero, the algorithm fails.

If  $f'(x)$  is very small then  $x_1$  will be very far from  $x_0$ .



# Problems with Newton's Method

- Cycle -



The algorithm cycles between two values  $x_0$  and  $x_1$

# Newton's Method for Systems of Non Linear Equations

*Given:*  $X_0$  an initial guess of the root of  $F(x) = 0$

*Newton's Iteration*

$$X_{k+1} = X_k - [F'(X_k)]^{-1} F(X_k)$$

$$F(X) = \begin{bmatrix} f_1(x_1, x_2, \dots) \\ f_2(x_1, x_2, \dots) \\ \vdots \end{bmatrix}, \quad F'(X) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \vdots \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \\ \vdots & & \end{bmatrix}$$

# Example

▣ Solve the following system of equations:

$$y + x^2 - 0.5 - x = 0$$

$$x^2 - 5xy - y = 0$$

Initial guess  $x = 1, y = 0$

$$F = \begin{bmatrix} y + x^2 - 0.5 - x \\ x^2 - 5xy - y \end{bmatrix}, \quad F' = \begin{bmatrix} 2x - 1 & 1 \\ 2x - 5y & -5x - 1 \end{bmatrix}, \quad X_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

# Solution Using Newton's Method

Iteration 1 :

$$F = \begin{bmatrix} y + x^2 - 0.5 - x \\ x^2 - 5xy - y \end{bmatrix} = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}, \quad F' = \begin{bmatrix} 2x-1 & 1 \\ 2x-5y & -5x-1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & -6 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 2 & -6 \end{bmatrix}^{-1} \begin{bmatrix} -0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 0.25 \end{bmatrix}$$

Iteration 2 :

$$F = \begin{bmatrix} 0.0625 \\ -0.25 \end{bmatrix}, \quad F' = \begin{bmatrix} 1.5 & 1 \\ 1.25 & -7.25 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 1.25 \\ 0.25 \end{bmatrix} - \begin{bmatrix} 1.5 & 1 \\ 1.25 & -7.25 \end{bmatrix}^{-1} \begin{bmatrix} 0.0625 \\ -0.25 \end{bmatrix} = \begin{bmatrix} 1.2332 \\ 0.2126 \end{bmatrix}$$

# Example

Try this

▣ Solve the following system of equations:

$$y + x^2 - 1 - x = 0$$

$$x^2 - 2y^2 - y = 0$$

Initial guess  $x = 0, y = 0$

$$F = \begin{bmatrix} y + x^2 - 1 - x \\ x^2 - 2y^2 - y \end{bmatrix}, \quad F' = \begin{bmatrix} 2x - 1 & 1 \\ 2x & -4y - 1 \end{bmatrix}, \quad X_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

# Example

Solution

| <i>Iteration</i> | 0                                      | 1                                       | 2                                           | 3                                                 | 4                                                 | 5                                                 |
|------------------|----------------------------------------|-----------------------------------------|---------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| $X_k$            | $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ | $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$ | $\begin{bmatrix} -0.6 \\ 0.2 \end{bmatrix}$ | $\begin{bmatrix} -0.5287 \\ 0.1969 \end{bmatrix}$ | $\begin{bmatrix} -0.5257 \\ 0.1980 \end{bmatrix}$ | $\begin{bmatrix} -0.5257 \\ 0.1980 \end{bmatrix}$ |